



Rugged Box (Parametric)



Whity

[VIEW IN BROWSER](#)

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Summary

Fully parameterized rugged tool box

[Hobby & Makers](#) > [Organizers](#)

Tags: [parametric](#) [water](#) [box](#) [organizer](#) [case](#) [storage](#)
[storagebox](#) [toolbox](#) [seal](#) [organize](#) [ruggedcase](#) [rugged](#)
[watertight](#) [parameterized](#) [parameterizable](#)

The design is based on the "[Tool box parametric](#)" by [Yanev](#) and provides some more options to parameterize it in Fusion 360.

Support my work

For non-Prusa parts and filaments, check out 3DJake ([International](#), [UK](#), [Germany](#), [France](#)).

I'll get a small cut of any order you fulfil using these links (at **no** additional cost for you 😊).

If you want to support my work directly, you can [buy me a coffee](#).

Also check out [my other models](#).

Remixes

Some of my own remixes:

- [Modular Slots - Shallow height \(single row version\)](#)
- [Modular Slots - Shallow height \(multi row version\)](#)
- [Modular Slots - Normal height](#)
- [Horizontal Inserts](#)
- [GoPro 9/10/11 Case](#)
- [GoPro 8 Case](#)
- [Brother TZe Label Tape Cartridges Box](#)
- [18650 Battery Storage Box for 50 Cells](#)
- [Box for Drill Handle and Collets](#)
- [Box for Radius Gauges](#)
- [Revopoint Mini Case](#)
- [Fluke Multimeter Case](#)
- [Dcreate Gear30 Spot Welder](#)
- [GEPRC TinyGO 4K Drone](#)
- [DMT 6" Diamond Whetstones & Holder Storage Box](#)
- [Filament Swatches Storage Box](#)

Important

The current STL files in the download section are all V1 only. If you want the newer design, use the Fusion 360 file and customize it as needed (that's recommended anyway, as there are many parameters to adjust)

Screws

I like these screws which have a nice round head:

[Allen hex socket button head screw](#)

Or, if you want no screw head at all, which I personally like even more:

[Hex socket set screws](#)

Note: I recommend the stainless steel ones. The blackened ones are oily and have to be cleaned.

Updates

24.08.2022:

- V2 release:
 - The front and back ribs are changed a bit to not have the full depth for the full height (see V2 details below). Looks much nicer like this (but V1 is still available)
 - The offset of the chamfer for each of them is configurable individually (...RibChamferOffset Parameters). This way you

can make the latch easier to open by lowering the offset, or make it less likely to open by accident by increasing the offset and cover the latch completely.

- If you wish you can now increase the size of the hinge to make it more sturdy (but I would not recommend it. It's more a hack...)
- The width of the outer parts of each hinge can now be set to make it more sturdy (only useful when using screws longer than 25 or 30 mm)

18.08.2022:

- Added "FLUKE Edition"
 - **28.08.2022:** Moved to a separate model: <https://www.printables.com/model/267524-fluke-rugged-case-with-foam-insert>

Changes made

- All parts in one F360 file
- Contains a seal to print in TPU (will not make it 100% water tight)
- Additional parameters:
 - The position of the latches and ribs (distance from the sides)
 - The diameter and length of the screw (to support different sizes / latch widths)
 - The inner chamfer size
 - Separate parameters for the widths of the ribs on the side and for the hinges/latches
 - Several tolerances to fine tune it for your printer

To assemble it you need 6 screws in total. For the provided STL files you will find the correct size in the folder names.

V2 Release

Here you see the main design change of the second version (V1 on the left, V2 on the right):



This remix is based on



Tool box parametric
by Yanev

Model files



Fusion 360 files

2 files



rugged-box-parametric-v2.f3d

☐ Go to "MODIFY" => "Change Parameters" to customize it



rugged-box-parametric-v1.f3d

☐ Go to "MODIFY" => "Change Parameters" to customize it



Size 120 x 70 x 40 (Screw M3x30) (V1 Design!)

4 files



rugged-box-box-120x70x40.stl



rugged-box-lid-120x70x40.stl



rugged-box-latch-120x70x40.stl



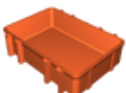
rugged-box-seal-120x70x40.stl

☐ OPTIONAL



Size 150 x 100 x 50 (Screw M3x30) (V1 Design!)

4 files



rugged-box-box-150x100x50.stl



rugged-box-lid-150x100x50.stl



rugged-box-latch-150x100x50.stl



rugged-box-seal-150x100x50.stl

☐ OPTIONAL



Size 200 x 185 x 60 (Screw M3x30) (V1 Design!)

4 files



rugged-box-box-200x185x60.stl



rugged-box-lid-200x185x60.stl



rugged-box-latch-200x185x60.stl



rugged-box-seal-200x185x60.stl

☐ OPTIONAL



STEP files

2 files



rugged-box-parametric-v2.step



rugged-box-parametric-v1.step

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